ABC Beverage Company: PH Compliance & Quality Enhancement Summary

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Executive Summary

Based on new regulatory criteria, ABC Beverage must demonstrate a thorough awareness of the elements that impact pH during the production process. Fluctuations in pH can affect product safety, taste, and shelf life. This paper presents analytical research carried out by our Data Science Team utilizing historical production data. The goal was to identify critical process factors that influence pH and create a prediction model that can estimate pH using real-time production inputs.

This project promotes compliance, increases process control, and improves overall product quality.

Purpose

The goal of this initiative was to:

1. Determine which manufacturing process factors have a major impact on pH levels.
2. Create a predictive model that estimates pH using production inputs.
3. Provide production managers with information and tools for maintaining ideal pH levels and ensuring regulatory compliance.

Data Overview

The dataset utilized in this investigation included 2,571 production records, each with 33 variables that included process parameters and product attributes. These metrics comprised pressure, temperature, fluid flow, chemical composition, and machine settings. A categorical variable, Brand Code, identified the product kind (1,2,3, or 4). The goal variable for modeling was pH, which is an important indicator of beverage stability and quality.

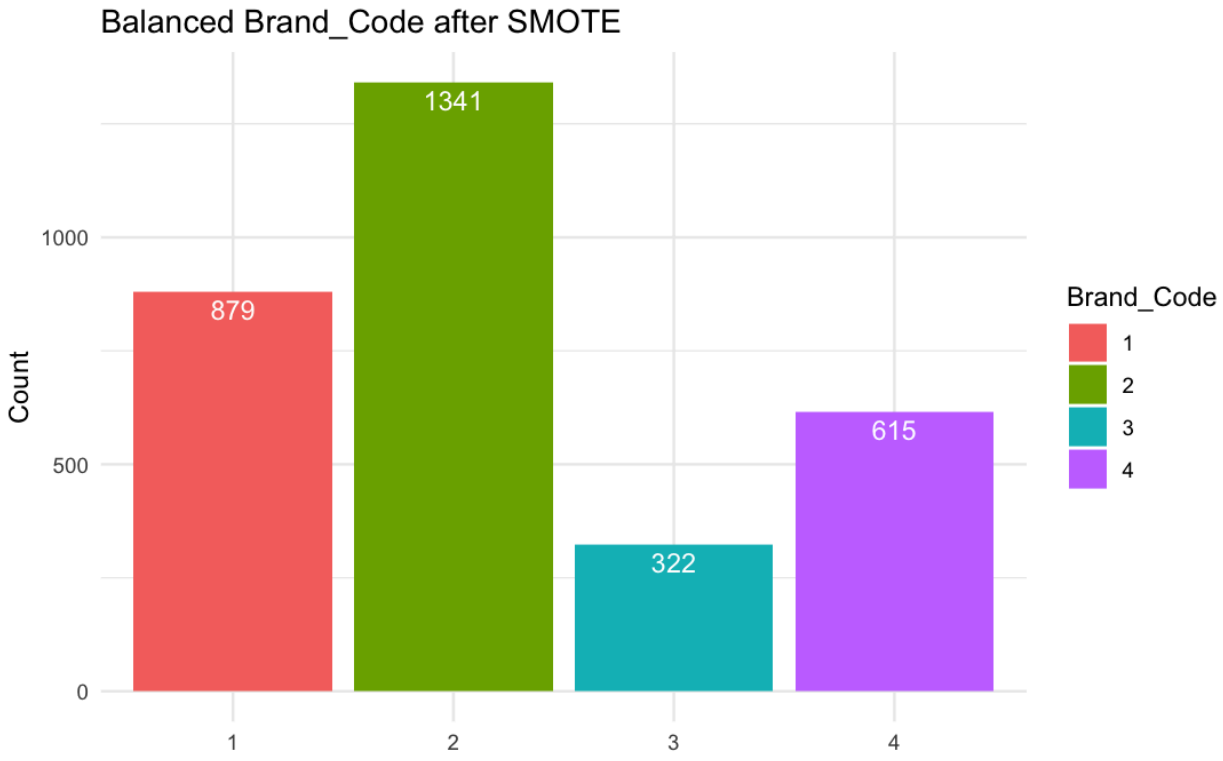
After cleaning the dataset, we found:

- Over 99% of data values were present, and missing values replaced with a value based on k-Nearest Neighbors imputation (kNN) imputation, which uses the mean value of the most similar **k** number of rows.
- Most columns were numerical and well suited for modeling.
- Brand B appeared more frequently than other brands, creating an imbalance that was adjusted during modeling.

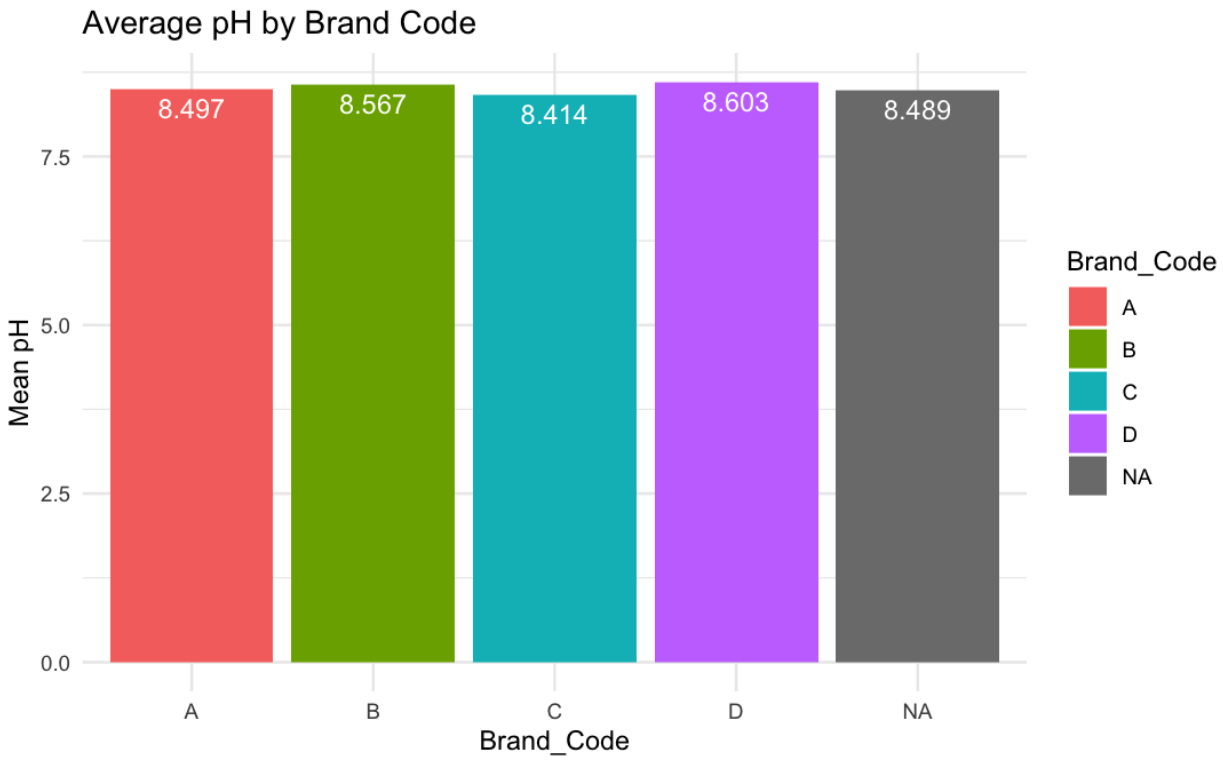
Background

We conducted a statistical analysis of the data to maximize accuracy of results for this assessment. This included checking for missing values, outliers that might skew the accuracy of data, and then replacing them with kNN values across each category. More details are listed in our technical report.

As stated above, our initial research labeled Brand B as the most frequently produced product, with over 1341 counts making it the highest among all product lines. Initial counts were slightly lower, but to adjust for underrepresented brands and balance the dataset, we used a process called SMOTE (Synthetic Minority Over-sampling Technique). We noted that Brand B could influence our predictive model for pH due to its high representation. A key factor in our analysis was to assess whether pH behavior differs by brand and, if so, account for it in our model modeling or stratify analysis accordingly.



Below are the average pHs by brand code.



Method Overview

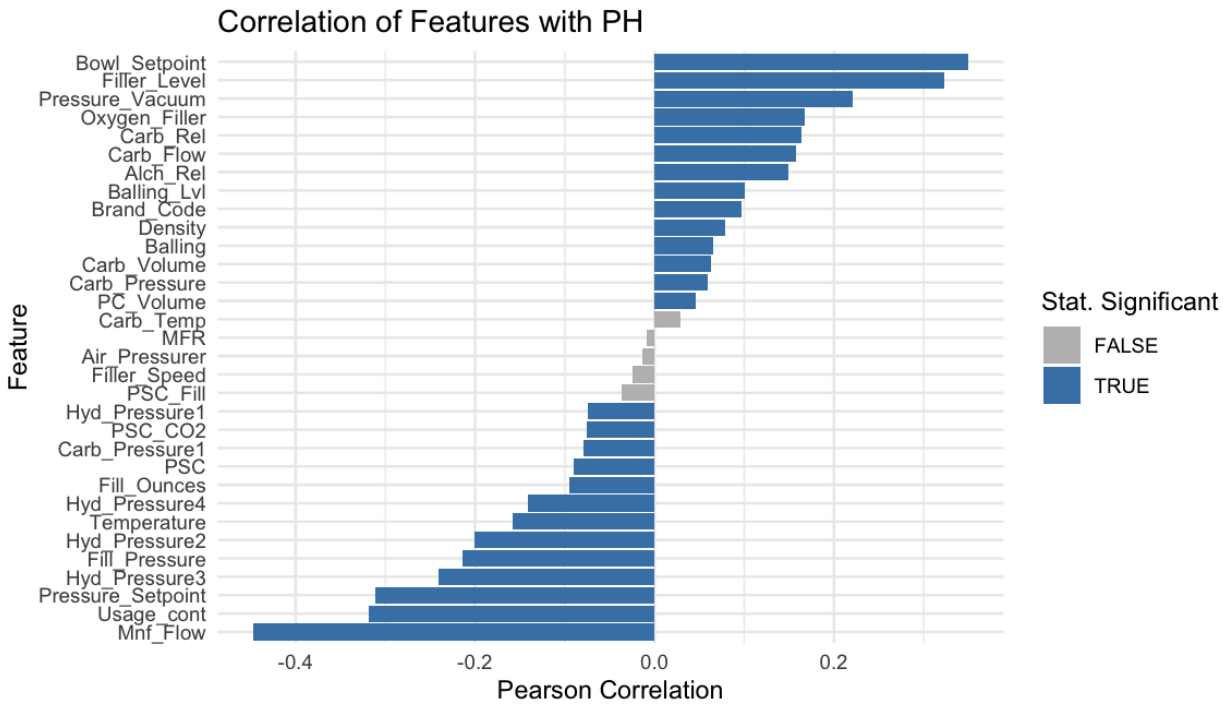
Next we checked for correlation between PH and other variables to assess whether they had a significant impact (raised the pH of the beverage), no impact, or negative impact (lowered the pH). The top three factors shown to have a significant impact on pH were:

- **Bowl Setpoint:** There was the largest positive link between pH and this setting, which regulates filling pressure or flow.
- **Filler Level:** The filling tank's liquid level was another reliable indicator.
- **Pressure Vacuum:** The parameters of the vacuum during bottling also had the third highest impact on pH.

Negative Correlation

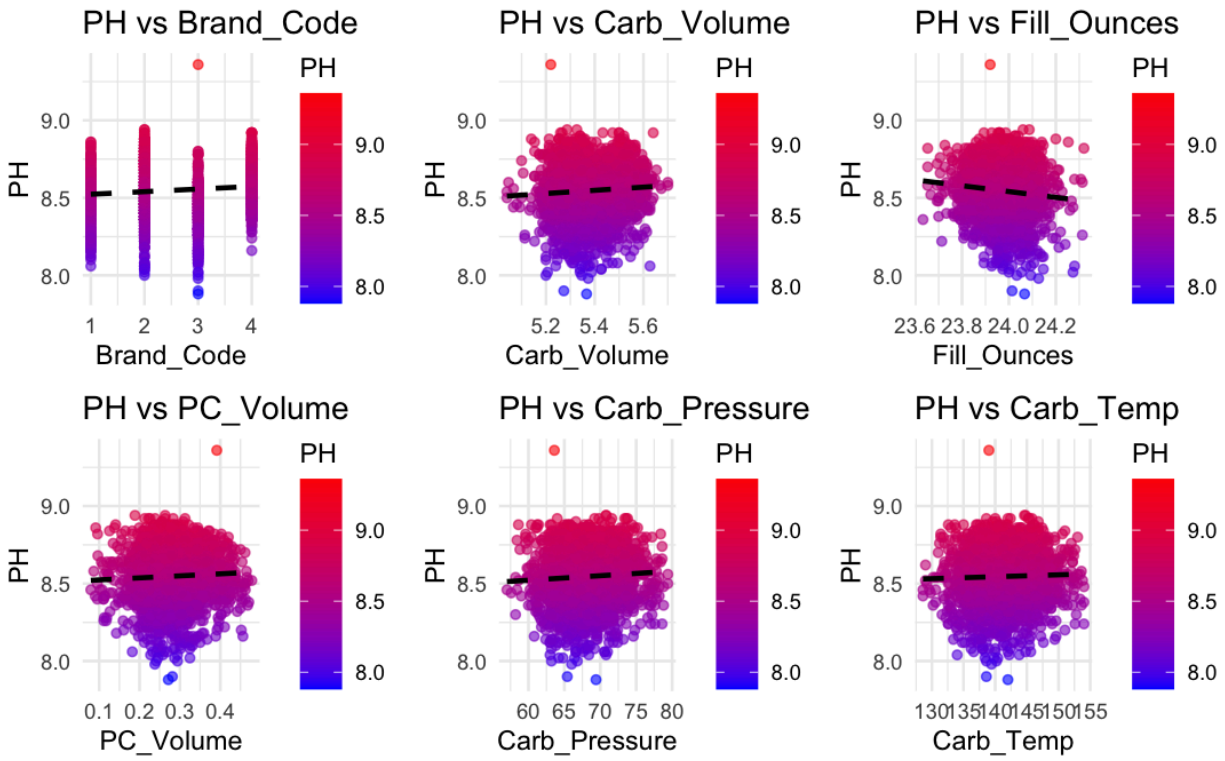
- **MnF Flow, Usage_cont, and Pressure_Setpoint:** These flow and usage metrics showed the highest negative correlation with pH.

Given their weaker correlations with pH, variables like air pressure, carb temperature, and filler speed could be deprioritized in operational monitoring. They are listed as “false” in the chart below.

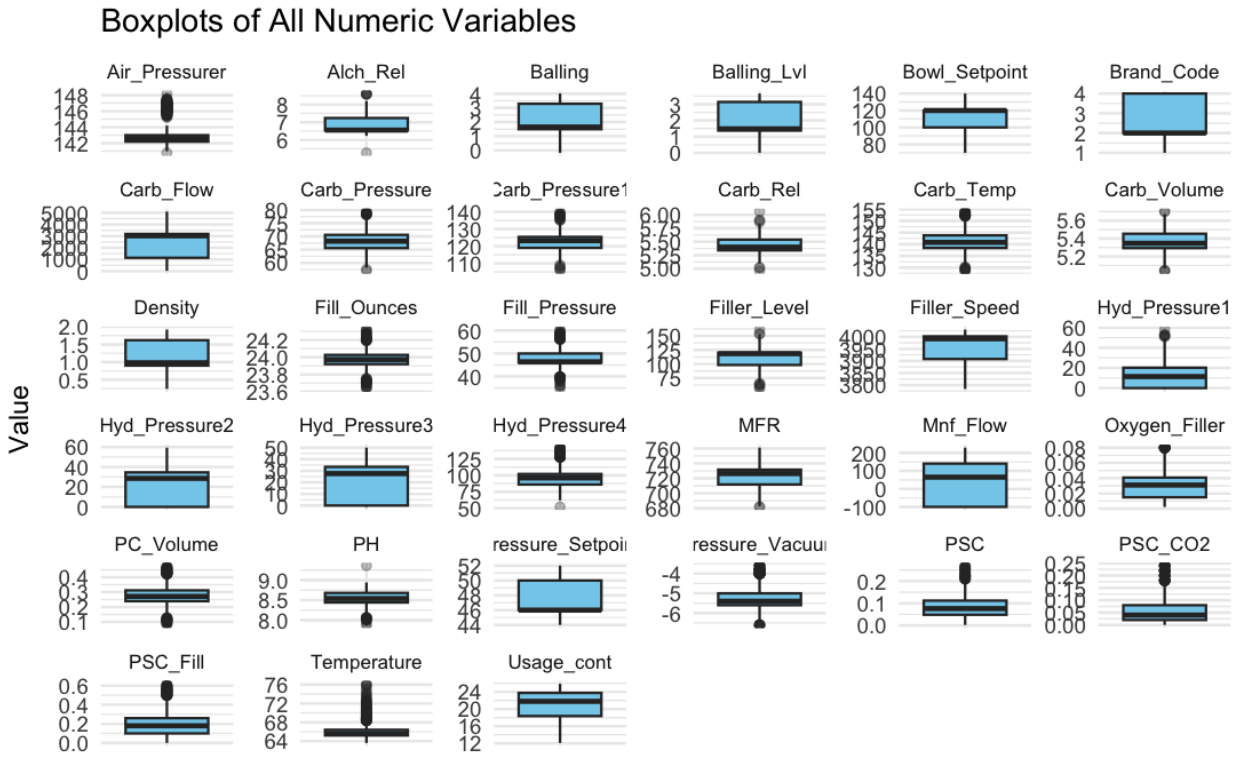


We also looked at the distributions of process variables to assess which processes appeared stable and provided consistent results, and in contrast with processes that likely needed improvement.

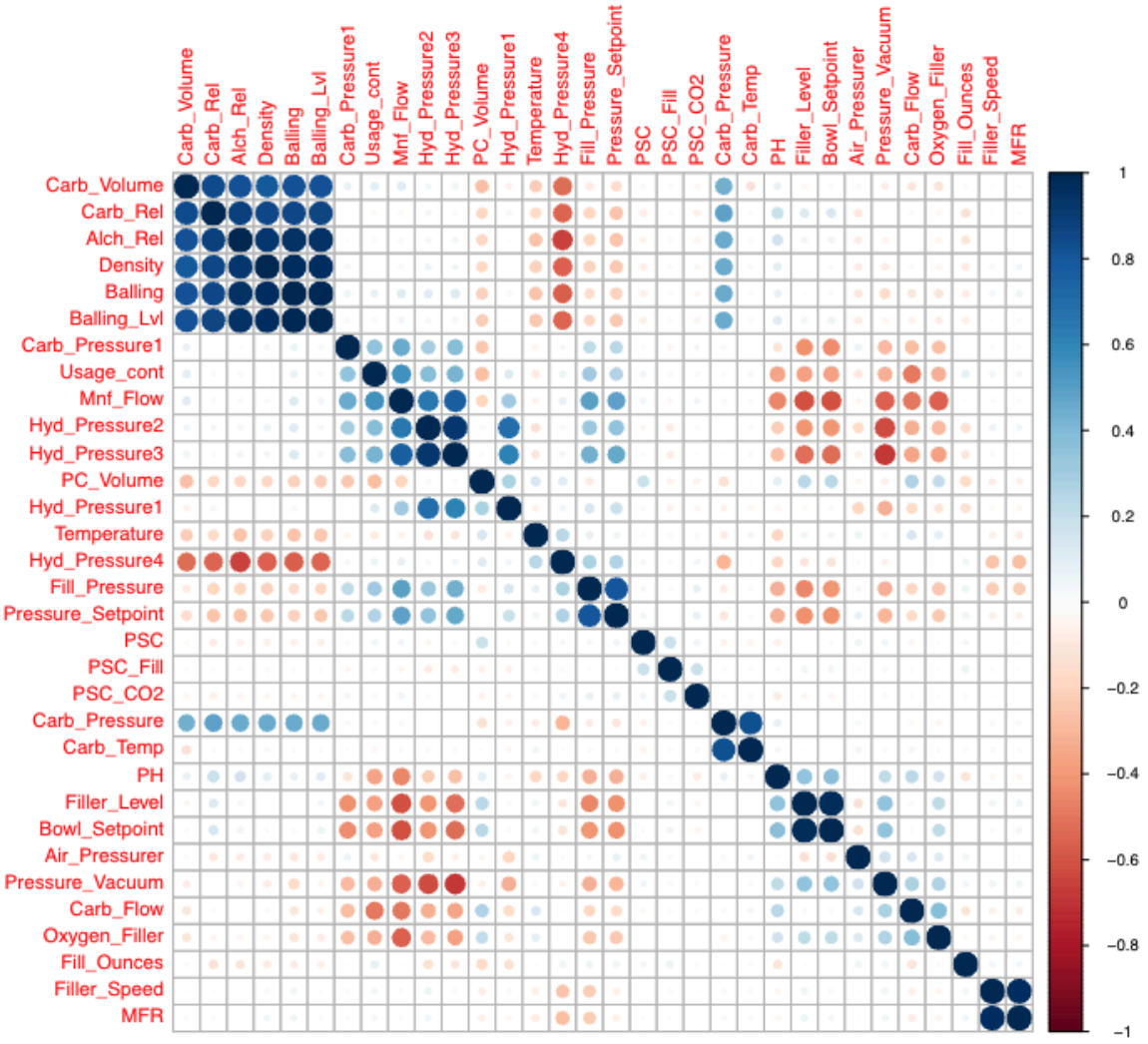
Some features, such as **Carb Pressure**, **Fill Ounces**, and **PC Volume**, display near-normal distributions, indicating stable process control. Others, such as **Oxygen Filler**, **PSC CO2**, **Carb Flow**, and **Usage Cont,** are heavily right-skewed, with long tails suggesting sporadic high values or potential outliers – processes that could be highly unpredictable and potentially needed better monitoring.



A second assessment below used a boxplot grid to highlight the distribution and outliers across all numeric variables of values the interquartile range (IQR), which is values between the 25-75% percentile. The black line in the box represents the median value, while the whiskers extend to the lowest and highest values within 1.5x IQRs of the box. Any other dots represent outliers.



We then assessed which variables were most correlated with others, and thus might lead to redundancy in our model.



In a final step, we perform a pre-training model correction of the data to remove extreme outliers that might skew the process.

Key Findings

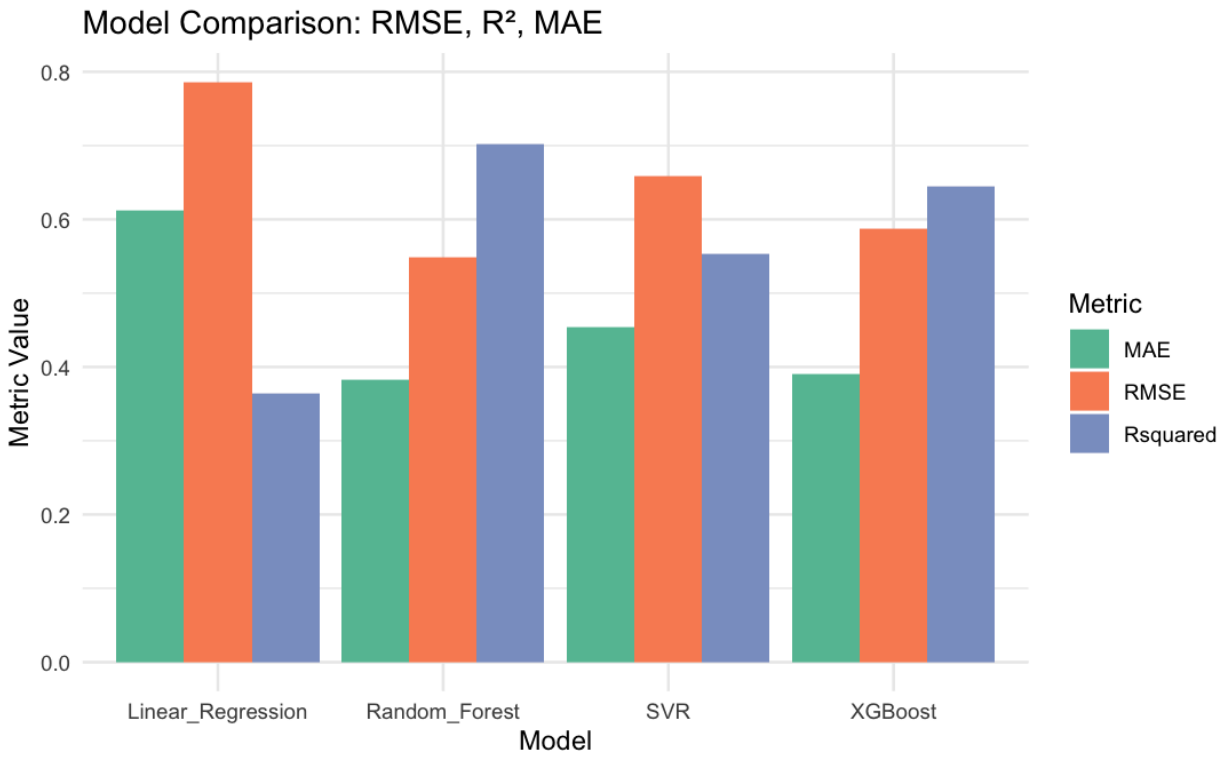
Modeling Results

An initial step was to build models to get an idea of what methods would perform best in predicting pH. Four predictive models to predict pH from production data were put to the test:

- Linear Regression
- Support Vector Machines (SVM)
- Random Forest
- Gradient Boosting Machine (XGBoost)

The prediction power and accuracy of the models were assessed. Below is a summary of the findings:

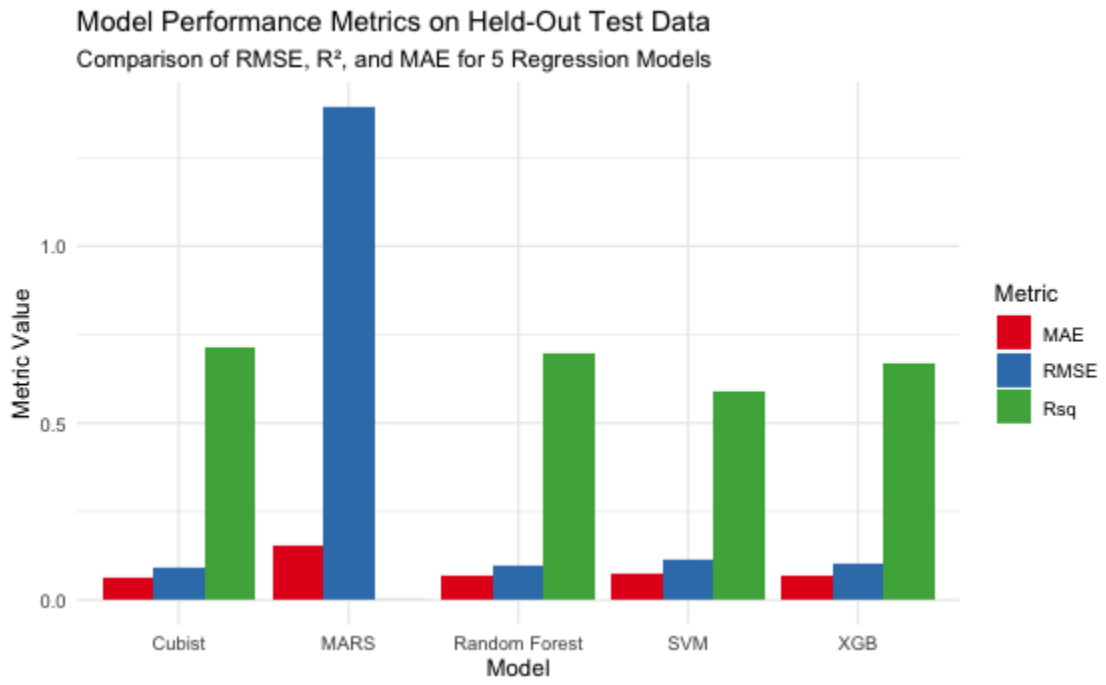
Model	Root Mean Square Error (RMSE)	R-squared (Accuracy)	Mean Average Error (MAE)
Linear Regression	0.79	0.364	0.612
Support Vector	0.69	0.554	0.454
Random Forest	0.55	0.703	0.383
XGBoost	0.59	0.645	0.391



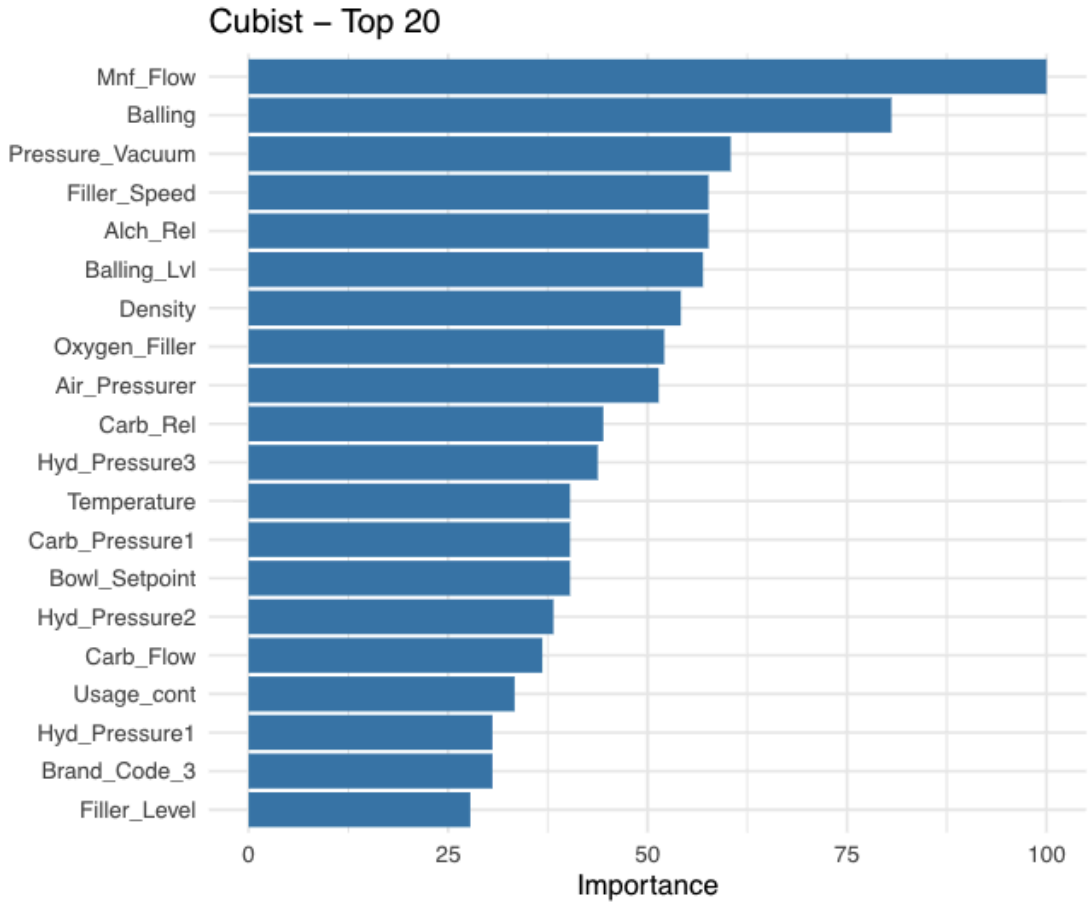
As expected the linear model had the poorest performance based on the lowest Rsquared value, which measures how well a regression fits the data, and the highest root mean squared error (RMSE), which quantifies the difference between predicted, and actual values.

Non-linear models were then selected to move forward with the aim of improving accuracy using cross validation and hyper-parameter tuning. The final set of models tested were **MARS**, **SVM**, **Random Forest**, **XGBoost,** and **Cubist.** The model with the highest R_squared value was used to predict pH for the new observations.

Model	Root Mean Square Error (RMSE)	R-squared (Accuracy)	Mean Average Error (MAE)
MARS	1.39	0.000614	0.153
SVM	0.112	0.588	0.076
Random Forest	0.099	0.695	0.069
XGBoost	0.101	0.669	0.070
Cubist	0.093	0.717	0.060



We ultimately selected the Cubist model, which is a rule-based regression model that converts a tree into a set of rules, with each rule associated with a linear model. Below is a list of the top 20 factors that most influenced pH.



Recommendations

1. **Track Important Variables:** Concentrate production supervision on the three most important factors: **Mnf_Flow**, **Balling**, and **Pressure_Vacuum**.
2. **Use Predictive Model:** The final Cubist model's pH predictions for the evaluation dataset are included in the Excel file that is given.
3. **Integrate Model into Operations:** To notify operators when the expected pH deviates from acceptable levels, the model may be integrated into dashboards or real-time systems.
4. **Training:** Production workers will receive quick training sessions to make sure they know which machine settings have the most effects on pH.

Deliverables

- **pH Values:** contains a spreadsheet format of predicted pH values for the dataset with link here
- **Technical Report**: constructing models, assessing performance, and writing code (included in .rmd file)

Conclusion

To conclude, The ABC Beverage dataset's exploratory study demonstrates a complex relationship between process variables and pH results, a crucial quality indicator in the beverage production industry. The distribution of variables predicts which processes are stable, and which are unpredictable and thus need more process control.

While the majority of variables exhibit minimal missingness and stable distributions, a few features—such as **PSC CO2**, **Hydraulic Pressure**, and **Usage Cont**—show substantial skewness and outliers, suggesting that they might be candidates for process improvement or sensor recalibration.

Both correlation coefficients and scatter plots support stronger linear relationships between pH and Bowl Setpoint, Filler Level, and Pressure Vacuum versus other predictors. On the other hand, factors like air pressure and filler speed have little predictive value, which might make it possible to simplify the model without noticeably sacrificing accuracy.

It's interesting to note that Brand Code's categorical imbalance, in which Brand B predominates the dataset, raises questions regarding possible modeling bias. Testing if pH behaves differently between brands is essential, particularly when applying the model to diverse production lines. To prevent skewed performance, brand-stratified sample or brand interactions should be introduced in future modeling efforts.

Overall, the dataset was suitable for machine learning applications and needed minimal preprocessing and imputation. To find actionable levers for keeping pH under specification, strong prediction models are built. The most important features were identified which provided the insights needed accurately predict pH..